

Professional Summary

Computing Science graduate and Data Engineer building data applications, analytics automation, and responsible GenAI workflows in the public sector. Experienced with Python, Databricks, CI/CD, data pipelines, and containerised systems; keen to build reliable data platforms for financial services.

Work Experience

GovTech | Singapore

Jun 2026 – Present

Data Engineer, Internal Audit

- Develop Databricks applications and technical proof-of-concepts that automate audit workflows and improve transparency of audit information for business users.
- Built an audit milestone tracking solution using Medallion Architecture to structure data flows across processing layers, with Databricks Asset Bundles integrated with GitLab for version-controlled development and deployment.
- Developing a Vite application that enables users to upload files to Databricks Volumes, trigger processing jobs, and access analytics dashboards; deployed on Northflank.

GovTech | Singapore

Sep 2025 – Apr 2026

Data Scientist Intern, Internal Audit

- Built and maintained an end-to-end analytics pipeline automating audit evidence processing for internal stakeholders.
- Developed **docXtract**, a modular Python tool that extracts embedded evidence from Office and Outlook files.
- Implemented iterative nested-file extraction across OOXML, legacy OLE, and MSG files, with provenance tracking, binary/content-aware deduplication, and password-protection detection.

Selected Projects

AuditSense – Capstone, GovTech Internal Audit

Sep 2025 – Mar 2026

Python | OOXML/CFB Parsing | Named Entity Extraction | TF-IDF | Knowledge Graph

- Designed a deterministic, provenance-preserving pipeline making AI-assisted audits traceable and reproducible.
- Evidence recovery layer (docXtract) extracted 213 nested artefacts up to 3 levels deep in 0.48s, deduplicating to 15 unique files (9.8 MB saved) with 100% precision and recall on a controlled reference corpus.
- Built a knowledge graph linking extracted entities to source chunks and documents for post-hoc grounding of chatbot workpapers, reporting entity match rate (69.2%) and flagging unmatched entities for reviewer follow-up.

Distributed Needy Fund Application System (DNFAS)

ML & LLM Service

NVIDIA NIM | Llama 3.1 8B Instruct | gRPC | Docker | Kubernetes

- Developed an experimental ML/LLM microservice for automated financial-aid application summarisation and 1–5 priority scoring using financial and employment indicators.
- Integrated NVIDIA NIM and Llama 3.1 8B Instruct into a containerised gRPC service on Docker and Kubernetes.

Education

Singapore Institute of Technology | University of Glasgow

Aug 2023 – Apr 2026

BSc. Computing Science

GPA: 4.64 / 5.0

Relevant Modules: Machine Learning, Data Analytics, Cloud Computing

Ngee Ann Polytechnic

Apr 2018 – Apr 2021

Diploma in Financial Informatics / Specialisation in Financial Analytics

GPA: 3.48 / 4.0

Awards / Achievements

- HackRift Champion [GovTech x NCS x SIT] (Dec 2024) – ScamSlayer POC
- Dell Tanzu Champion (Apr 2025) – Grub microservices project
- Provost List AY2024/25

Technical Skills

- Languages & Data: Python, Java, C, SQL, R, Databricks, TensorFlow, Machine Learning, LLM Integration
- Data Engineering & DevOps: Medallion Architecture, Databricks Asset Bundles, GitLab CI/CD, Git
- Cloud & Distributed Systems: Docker, Kubernetes, gRPC, Vite, Northflank